350 × 350

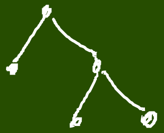
$$\left(\boxed{A} \right) \in M_{m,n}(\mathbb{R}) \qquad m \cdot n$$

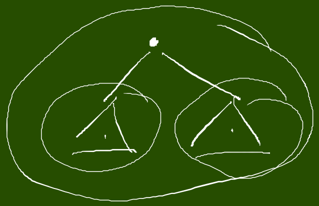
$$m \underbrace{\begin{pmatrix} U \end{pmatrix}}_{m} \overset{\Sigma}{\underbrace{\begin{pmatrix} \sigma_1 & & 0 \\ m & \sigma_2 \ddots & \\ 0 & n & \end{pmatrix}}} \underbrace{\begin{pmatrix} V^t \end{pmatrix}}_{n} n$$

$$\begin{cases} U^{-1} = U^t \\ V^{-1} = V^t \end{cases} \qquad \sigma_1 \geq \sigma_2 \geq \dots \geq 0$$

SVD

Singular value decomposition





Vectori

$v = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$ $(\mathbb{K}, +, \cdot, 0, 1)$ $\mathbb{R}, \mathbb{C}, \mathbb{Z}_p$ (p prim)

$\mathbb{R}^n, \mathbb{C}^n, \mathbb{Z}_p^n$

$\underset{\equiv}{\mathbb{K}^n} = \left\{ v \in \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \middle| x_1 \dots x_n \in \mathbb{K} \right\}$ $\mathcal{M}_{n,1}(\mathbb{K})$